PROGRAMME : Diploma Programme in Information Technology/Computer Technology
COURSE : Software Engineering(SWE) **COURSE CODE :** 232146

LEARNING AND ASSESSMENT SCHEME:

Learning Scheme					Credits	Assessment Scheme											Total Marks
Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL and TSL				Based on Self Learning		
											Practical						
CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
							Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
03	00	02	01	06	03	03	30	70	100	40	25	10	--	--	25	10	150

IKS Content: Total Learning Hours for Term: 00 Hrs
# Indicates Online Examination, @ Internal assessment

1.0 RATIONALE:

The Software Engineering curriculum provides a comprehensive understanding of software development methodologies and best practices. It covers essential topics such as system analysis, design, coding, testing, and project management. This approach ensures students can effectively create, maintain, and manage software projects. Emphasizing practical experience, it prepares students to tackle real-world challenges and adapt to industry standards. By integrating theoretical knowledge with hands-on skills, the syllabus aims to produce competent professionals ready for the evolving software industry.

2.0 INDUSTRY / EMPLOYER EXPECTED OUTCOME:

The aim of the course is to help the student to attain the following industry /employer expected outcome: students will be well-prepared for careers in software development, with a solid understanding of fundamental Software Engineering.

3.0 COURSE OUTCOMES:

The course content should be taught and learning imparted in such a manner that students are able to acquire required learning outcome in cognitive, psychomotor and affective domain to demonstrate following course outcomes:

1. Select suitable Software Process model for software development.
2. Prepare software requirement specifications.
3. Use Software modelling to create data designs.
4. Estimate size and cost of Software product.
5. Apply project management and quality assurance principles in Software development.

4.0 COURSE DETAILS:

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
Unit-I **Software Development Process**	1a. Suggest the attributes that match with standards for the given software application. 1b. Recommend the relevant software solution for the given problem with justification. 1c. Select the relevant software process model for the given problem statement with justification. 1d. Suggest the relevant activities in Agile Development Process	1.1 Software, Software Engineering as layered approach and its characteristics, Types of software. 1.2 Software development framework. 1.3 Software Process Model: waterfall model 1.4 Incremental Process Models: RAD model. 1.5 Evolutionary Process Model: Prototyping model, Spiral	08

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
	in the given situation.	model. 1.6 Agile Software development: Agile Process and its importance, Extreme Programming, Adaptive Software Development (ASD), Scrum, Dynamic Systems Development Method (DSDM), CRYSTAL. Agile Unified Process (AUP). 1.7 Selection criteria for process model.	
Unit-II **Software requirement Engineering**	2a. Apply the principles of software engineering for the given problem. 2b. Choose the relevant requirement engineering steps in the given problem. 2c. Represent the requirement engineering model in the given problem. 2d. Prepare SRS for the given problem.	2.1 Core Software Engineering Practices and its importance, Core principles. 2.2 Communication Practices, Planning Practices, Modelling practices, construction practices, software deployment (Statement and meaning of each principle for each practice). 2.3 Requirement Engineering: requirement Gathering and Analysis, Types of requirements (Functional, Product, organizational, External requirements), Eliciting requirements, Developing Use Cases, Building requirements models, requirements Negotiation, Validation. 2.4 Software requirements Specification: Need of SRS, Format, and its Characteristics.	09
Unit-III **Software Modeling and Design**	3a. Identify the elements of analysis model for the given software requirements. 3b. Apply the specified design feature for software requirements modelling. 3c. Represent the specified problem in the given design notation. 3d. Explain the given characteristics of software testing. 3e. Draw Use-case, Class Diagrams, Sequence Diagrams for software project.	3.1 Translating requirements model into design model: Data Modelling. 3.2 Analysis Modelling: Elements of Analysis model. 3.3 Design modelling: Fundamental Design Concepts (Abstraction, Information hiding, Structure, Modularity, Concurrency, Verification, Aesthetics). 3.4 Design notations: Data Flo Diagram (DFD), Structured Flowcharts, Decision Tables. 3.5 UML Modelling: Use-Case, Class Diagrams, Sequence Diagrams. 3.6 Testing – Meaning and purpose, Testing methods-	09

Unit	Major Learning Outcomes (in cognitive domain)	Topics and Sub-topics	Hours
		Black-Box and White –Box, Static and Dynamic testing, Levels of Testing, V- Model.	
Unit-IV **Software Project Estimation**	4a. Estimate the size of the software product using the given method. 4b. Estimate the cost of the software product using the given empirical method. 4c. Evaluate the size of the given software using COCOMO model. 4d. Apply the RMMM strategy in Identified risks for the given software development problem.	4.1 The Management spectrum- 4P's 4.2 Metrics for Size Estimation: Line of Code (Loc), Function Points (FP). 4.3 Project Cost Estimation Approaches: Overview of Heuristic, Analytical and Empirical Estimation. 4.4 COCOMO (Constructive Cost Model), COCOMO II. 4.5 Risk Management: Risk Identification, Risk Assessment, Risk Containment, RMMM strategy.	10
Unit-V **Software project Managemen t and quality assurance**	5a. Use the given Scheduling technique for the identified project. 5b. Draw the activity network for the task. 5c. Prepare the timeline chart /Gantt chart to track progress of the given project. 5d. Describe the given Software Quality Assurance (SQA) activity. 5e. Describe features of the given software quality evaluation standard.	5.1 Project Scheduling: Basic principles, work breakdown structure, Activity network 5.2 Scheduling techniques: Critical path Method (CPM), Program Evaluation review Techniques (PERT). 5.3 Project Tracking: Timeline Charts, Earned Value Analysis, Gantt Charts 5.4 Software Quality Management vs. Software Quality Assurance. Phases of Software Quality Assurance: Planning, Activities, audit, review. 5.5 Quality Evaluation standards: Six Sigma, ISO for software, CMMI: Levels, Process areas.	09

5.0 SUGGESTED SPECIFICATION TABLE WITH MARKS (THEORY):

Unit No.	Unit Title	Distribution of Theory Marks			
		R Level	U Level	A and above Levels	Total Marks
I	Software development process	04	04	04	12
II	Software requirement Engineering	02	04	08	14
III	Software Modelling and Design	02	04	08	14

Unit No.	Unit Title	Distribution of Theory Marks			
		R Level	U Level	A and above Levels	Total Marks
IV	Software Project Estimation	04	04	08	16
V	Software Project Management and quality assurance	04	04	06	14
	TOTAL	**16**	**20**	**34**	**70**

6.0 LABORATORY LEARNING OUTCOME AND ALLIED PRACTICAL/ TUTORIAL EXPERIENCES:

The tutorial/practical/assignment/task should be properly designed and implemented with an attempt to develop different types of cognitive and practical skills **(Outcomes in cognitive, psychomotor and affective domain)** so that students are able to acquire the desired programme outcome/course outcome.

Note: Here only outcomes in psychomotor domain are listed as practical/exercises. However, if these practical/exercises are completed appropriately, they would also lead to development of **Programme Outcomes/Course Outcomes in Affective domain** as given in the mapping matrix for this course. Faculty should ensure that students also acquire Programme Outcomes/Course Outcomes related to affective domain.

Sr. No.	Unit No.	Laboratory Learning Outcome	Laboratory Experiment Title	Hours
01	I	1a. Use Software tools to rite problem statement and identify scope of the project.	• Write problem statement to define the project title with bounded scope of the project.	02
02	I	2a. Use appropriate process model and activities related to project.	• Select relevant process model to define activities and related tasks set for assigned project.	02
03	II	3a. Apply the principles of requirement engineering.	• Gather application specific requirements for assimilate into RE(requirements engineering) model	02
04	II	4a. Create SRS document for the project.	• Prepare broad SRS(software requirement software) for the above selected project	02
05	II	5a. Construct use case diagram for software models.	• Prepare use-cases and draw use case diagram using software Modeling Tool.	02
06	II	6a. Design activity diagram for the project.	• Develop the activity diagram to represent flow from one activity to another for software development.	02
07	III	7a. Draw data flow diagram for the project 7b. Create Decision tables and E-r diagram.	• Develop data designs using DFDs (Data flow diagram), Decision tables and E-R (entity – relationship) diagram.	02
08	III	8a. Represent software project by Class Diagram.	• Draw class diagram, Sequence diagram, State Transition Diagram for the assigned project.	02
09	III	9a. Prepare decision table for the Project.	• Create decision table for a Project.	02
10	IV	10a. Identify risk involved in the Project. 10b. Prepare RMMM plan.	• Identify risks involved in the project and prepare RMMM (RMMM- Risk Management, Mitigation and	02

Sr. No.	Unit No.	Laboratory Learning Outcome	Laboratory Experiment Title	Hours
			monitoring) plan.	
11	IV	11a.Estimate size of the project using function point metric.	• Evaluate size of the project using Function point metric for the assigned project.	02
12	IV	12a.Estimate size of the project using COCOMO approach.	• Estimate cost of the project using COCOMO (Constructive Cost Model) / COCOMO II approach for the assigned project.	02
13	V	13a.Prepare project schedule using CPM/PERT technique.	• Use CPM (Critical Path method) / PERT (Programme Evaluation and review Technique) for scheduling the assigned project.	02
14	V	14a.Prepare SQA plan to ensure various quality processes.	• Prepare SQA plan that facilitates various attributes of quality of process.	02
15	V	15a.Prepare SQA plan to ensure various quality processes.	• Prepare SQA plan that facilitates various attributes of quality of product.	02
Total				30

7.0 SELF LEARNING: (Assignment / Activities for Specific Learning / Skill Development / Online Courses / Micro projects.

Other than the classroom and laboratory learning, following are the suggested Student-related co-curricular activities. This can be undertaken to accelerate the attainment of the various outcomes in the course.

Each student must perform at **least one activity**. Following are suggested as sample assignments. Faculty is expected to assign individual assignment to each student and keep record in hard/soft copy as required for assessment.

Micro project

- Apply the principles of software engineering for Portfolio website for showcasing Skills and Work , Searchability and Online Presence , Demonstrating Growth and Progress , Career Advancement and Networking and Prepare complete technical document.
- Apply the principles of software engineering for Chatbot Application to create an intelligent chatbot to enhance customer support processes, providing efficient and personalized assistance and Develop technical document.
- Apply the principles of software engineering for Online Chat Application Project that enables users to exchange messages and communicate with each other in real-time. It allows individuals or groups to have conversations, share information, and collaborate instantly over the Internet. Online Chat Application is designed to provide a responsive and interactive experience, where messages are delivered and displayed immediately as they are sent and Prepare Complete technical document.

Assignment

- Estimate Cost of software using any tool and risk involved in the library Management System
- Create DFDs, Activity Diagram, ER-Diagrams for Student Management System.
- Visit any medical shop and collect requirements from shop keeper and create SRS document

8.0 SPECIAL INSTRUCTIONAL STRATEGIES (If any):

These are sample strategies, which the teacher can use to accelerate the attainment of the various outcomes in this course,

1. Guide students in undertaking mini projects.
2. Demonstrate students thoroughly before they start doing the practice.
3. Encourage students to refer different website and YouTube channels to have deeper understanding of the subject.
4. Observe and monitor the performance of students in lab.
5. Use of online platforms like MOOCs, Swayam, Spoken tutorial, NPTEL, Infosys springboard etc.

9.0 ASSESSMENT METHODOLOGY

1. Formative assessment: TH
   Periodic Test: 30marks
2. Summative assessment: TH
   70 marks Final theory paper.
3. Formative assessment: PR
   Lab performance, Viva
4. Self learning includes micro projects /assignments/ other activities.

10.0 LEARNING RESOURCES:

A) Books:

Sr.No.	Author	Title of Book	Publication
1	Roger Pressman and Bruce Maxim	Software Engineering : A Practitioner's approach	McGraw Hill Education, 9th Edition, ISBN10: 1259872971 | ISBN13: 9781259872976
2	Richard Fairley	Software Engineering Concepts	McGraw Hill Education, ISBN-13 : 978-0074631218
3	Deepak Jain	Software Engineering :Principles and practices	Oxford University Press, New Delhi, ISBN-13 : 978-0195694840
4	Waman S. Jawadekar	Software Engineering-Principles and Practice	Tata McGraw Hill Publishing Co. Ltd.
5	Srinivasan Desikan	Software Testing	Pearson Education, ISBN-13 : 978-8177581218

B) Software/Learning Websites:

1. https://www.geeksforgeeks.org/software-engineering-introduct ion-to-software-engineering/
2. https://www.tutorialspoint.com/software_engineering/index.htm
3. https://www.sei.cmu.edu/
4. https://www.youtube.com/watch?v=WjwEh15M5Rw
5. https://app.diagrams.net/

C) Major Equipment/ Instrument with Broad Specifications:

Sr.No.	Equipment	Specification
1	Desktop Computer	PC Specifications to be followed: Processor: i3 or i5 (or Higher) RAM: 4 GB (or Higher) HDD: 1 TB SATA Monitor: TFT LCD

		OS: Genuine Windows 8 or 10 Professional or Home Premium or Windows 8 or 10 Ultimate Internet Connection
2	LCD Projector	Display Type: LCD Light Output: 3200 Lumens
3	Software	Any UML tool, :Open Project, Ganttproject 3.3, Draw.io, Decision Table Maker, Tiny tools, Projectriskmanager, Open source Software such as Jira.

11.0 MAPPING MATRIX OF PO's, CO's and PSO's:

Course Outcomes	**Programme Outcomes (PO's)**							**Programme Specific Outcomes (PSO's)**		
	1	**2**	**3**	**4**	**5**	**6**	**7**	**1**	**2**	**3**
CO1	L	M	M	M	L		L			
CO2	M	H	H	M	L		L			
CO3	M	M	M	H			L			
CO4	M	M	M	H		M	M			
CO5		H	M	H	L	M	M			

H: High Relationship, M: Moderate Relationship, L: Low Relationship.

12.0 SUGGESTED QUESTION PAPER PROFILE: